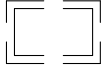
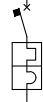
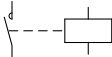
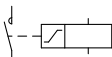
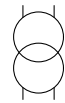


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TENDRARA PROJECT GAS TREATMENT AND LNG PRODUCTION PLANT ON RENTAL BASIS																					
SAFETY CRITICAL ELEMENT LIST																					
IFI	2	11/04/2025	AS-BUILT				ADF	GDP	IC												
IFI	1	31/03/2025	GENEERAL REVIEW				ADF	GDP	IC												
IFI	0	26/03/2024	ISSUED FOR COMMENTs				ADF	GDP	IC												
Status	Rev. n.	Rev. n.	Rev. n.				Rev. n.	Rev. n.	Rev. n.												
Rev. Index		Date	Description				Prepared	Checked	Approved												
Client							Project Title														
ManaEnergy Managem Group MANAENERGY Mahaj Riad Center, Batiment 6, 3eme etage Avenue Al Arz, Hay Ryad, 10100 Rabat, Morocco							TENDRARA PROJECT														
Contractor							Provision of a m-LNG plant for production for ManaEnergy														
ITALFLUID ITALFLUID GEOENERGY Marocco Branch Boulevard de Bourgogne Rue Jaafar Ibn Habib Residence Al Mashrik II 1ere Etage N3 Casablanca																					
Engineering																					
ITALFLUID ITALFLUID GEOENERGY Via Antonio Vivaldi, 4 Montesilvano (PE) Tel:085-4224374 / fax: 085-4220742																					
Facility Name			Location				Scale		Sheet of Sheets			esa esaelettromeccanica.com AMTIVO SO 9001 CERTIFIED INGEGNERIA ELETTRICA - QUADRI DI DISTRIBUZIONE E AUTOMAZIONE - CABINE DI TRASFORMAZIONE MT/BT – PLC – HMI – SCADA - ENERGIE RINNOVABILI CONTRADA TESINO, 47 - 63065 RIPATRANSONE (AP) – ITALY - TEL. +39 0735 90523 WWW.ESAELETTROMECCANICA.COM - INFO@ESAELETTROMECCANICA.COM - ESA@PEC.IT - P.IVA e C.F. 00768110447									
TENDRARA			MOROCCO				A4		1 of 27												
Document Title						Doc. Code															
P-PN-S221A.1						ELD01-ING-FR01															
LNG PUMP SKID S-221A						Vendor Doc. code IE24090_SE01_P-PN-S221A.1															
			Project : P–PN–S221A.1				Customer : ManaEnergy Managem Group				Object :										
General Contractor :		Manufacture : esa esaelettromeccanica.com		LNG PUMP SKID S–221A				Contract : IE24090				Rev. 2		Data : 11/04/2025				Date : 26/03/2024			
								S/N : IE24090_SE01				Rev. 1		Data : 31/03/2025				Sheet : 1 of : 27			

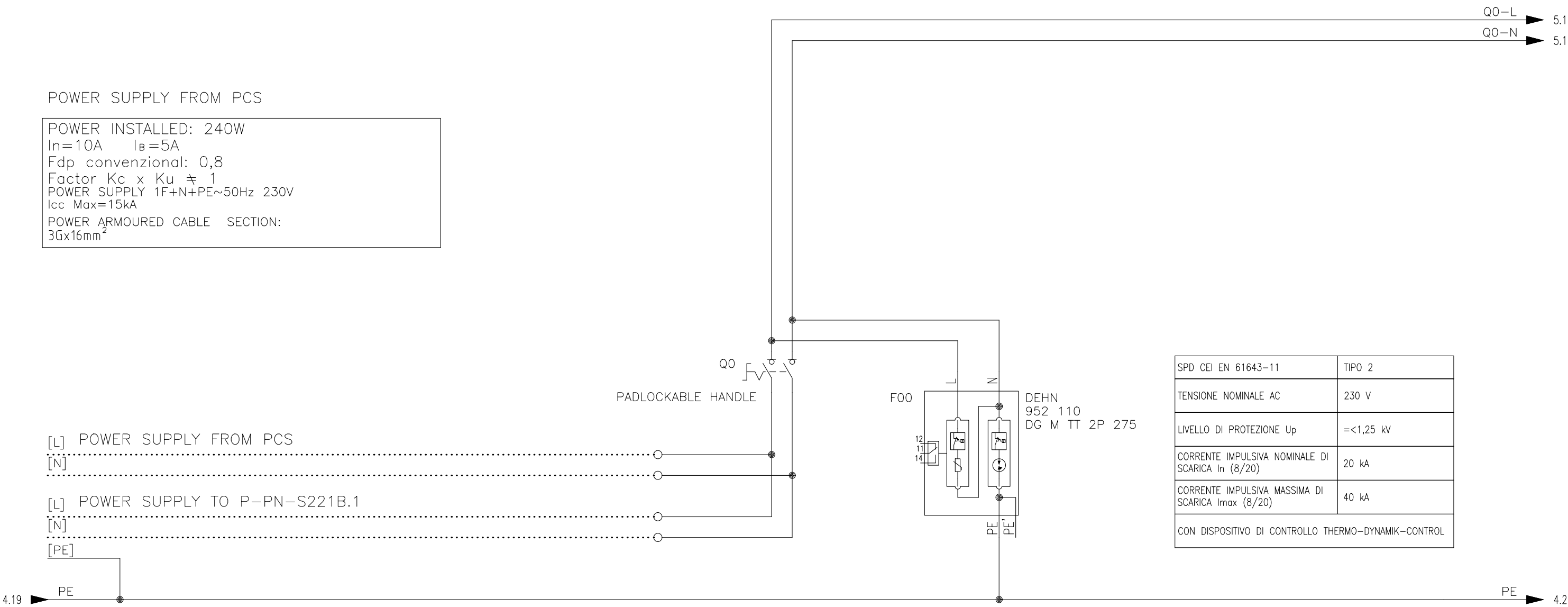
CIRCUIT SYMBOLS ACCORDING TO CEI STANDARD

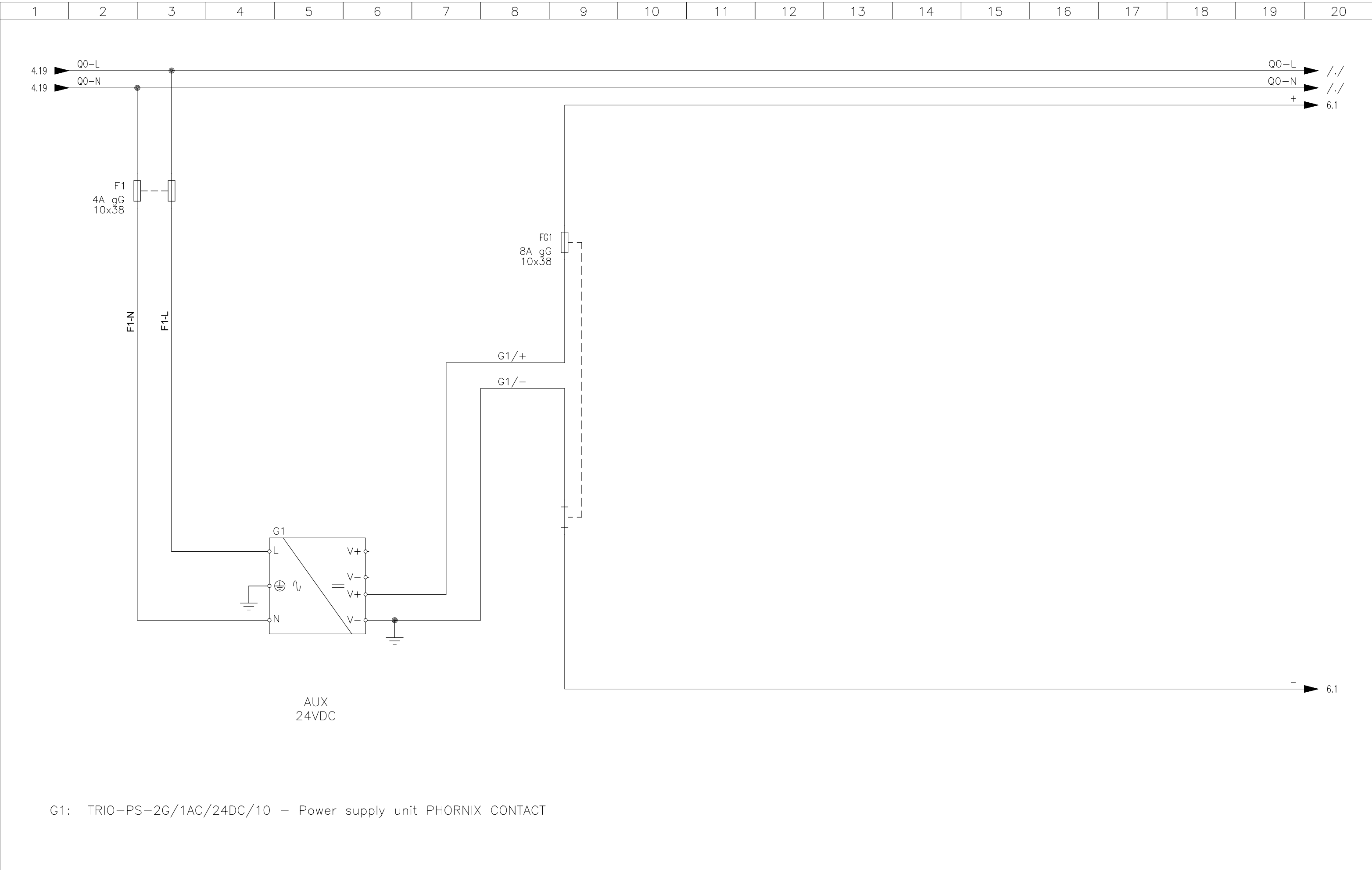
	ISOLATOR		ELECTRONIC TIMER WITH PICK-UP DELAY		MUSHROOM PUSH-BUTTON PULSE CONTACT WITH STAY-PUT MAKE CONTACT OPERATED BY STRIKING		STEDY OR FLASHING LIGHT
	FUSED ISOLATOR		ELECTRIC PULSE METER		KEY OPERATING DEVICE FOR SELECTORS, PUSH-BUTTON, ETC.		PLC UNIT
	CIRCUIT BREAKER WITH THERMAL AED MAGNETIC OVERCURRENT RELEASES		FUSE CARRIER		PRESSURE SWITCH		CONTROL INSTRUMENT
	CIRCUIT BREAKER WITH MAGNETIC RELEASE		MAKE CONTACT OF CONTACTOR		TEMPERATURE SWITCH		ENERGY METER
	CIRCUIT BREAKER WITH THERMAL, MAGNETIC AND DIFFERENTIAL RELEASES		MAKE CONTACT OF CONTACTOR		FLOW SWITCH		MEASURING INSTRUMENT
	RESIDUAL CURRENT CIRCUIT BREAKER "AC" TYPE FOR ALTERNATING CURRENT "A" TYPE FOR ALTERNATING OR PULSATING CURRENTS		MAKE AND BREAK CONTACT		FLOAT SWITCH		VOLTAGE TRANSFORMER
	THERMAL RELAY (OVERLOAD RELAY)		MAKE AND BREAK CONTACT WITH DROP-OUT DELAY		BUCHHOLZ RELAY		CURRENT TRANSFORMER
	COIL OF CONTACTOR RELAY		MAKE AND BREAK CONTACT WITH PICK-UP DELAY		CAM OPERATED DEVICE		ACOUSTIC SIGNALLING DEVICE
	COIL OF SHUT-OFF ALARM MAKE CONTACT ON ENERGIZATION		MAKE AND BREAK CONTACT WITH DROP-OUT AND PICK-UP DELAY (FLIP-FLOP)		PROXIMITY SWITCH		RESISTANCE
	COIL OF SHUT-OFF ALARM DENERGIZES IN CASE OF UNDERVOLTAGE		MAKE AND BREAK CONTACT OF THERMAL RELAY		TEMPERATURE SENSOR		CAPACITOR
	COIL OF OPERATING DEVICE WITH DROP-OUT DELAY		MAKE AND BREAK CONTACT OF LIMIT SWITCH		PHOTOELECTRIC SENSOR		SINGLE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH PICK-UP DELAY		MULTI POSITION SWITCH (GENERAL SYMBOL)		SOLENOID VALVE		THREE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH DROP-OUT AND PICK-UP DELAYS		MAKE AND BREAK CONTACT OPERATED BY TURNING WITH SPRING RETURN		PHOTOCELL		
	COIL OF LATCHING RELAY (STEP-BY-STEP FUNCTION)		MAKE AND BREAK CONTACT OPERATED BY PUSHING WITH SPRING RETURN				
	ELECTRONIC OR ELECTRO-MECHANICAL TIME SWITCH		MAKE AND BREAK CONTACT OPERATED BY PULLING WITH OR WITHUOT SPRING RETURN				

CONDUCTOR COLOR CODING INSIDE CONTROL PANEL	
AC AND DC POWER CIRCUIT	BLACK
NEUTRAL CONDUCTOR OF POWER CIRCUIT	LIGHT–BLUE
PROTECTIVE CONDUCTOR (EARTH CONDUCTOR)	YELLOW–GREEN
110/220V AC AUXILIARY CIRCUIT (*)	RED
24V AC AUXILIARY CIRCUIT	WHITE
24V AC AND DC PLC INPUT	GRAY
24V AC PLC OUTPUT	WHITE
24V AC AND DC INTERLOCKING CIRCUIT	ORANGE
24V DC AUXILIARY CIRCUIT AND PLC OUTPUT	DARK–BLUE
INTRINSIC SAFE SIGNAL	BLUE
(*) : RED FOR 110V AND BROWN FOR 220V, IN CASE BOTH VOLTAGE ARE USED	

CODE COLOUR	
COLOUR	CODE LETTERS
BLACK	BK
BROWN	BN
RED	RD
ORANGE	OG
YELLOW	YE
GREEN	GN
BLUE	BU
GREY	GY
WHITE	WH
SILVER	SR
CLEAR	CL
GREEN AND YELLOW	GNYE

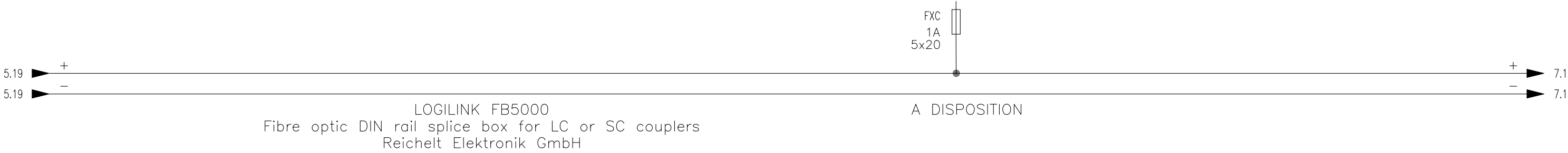
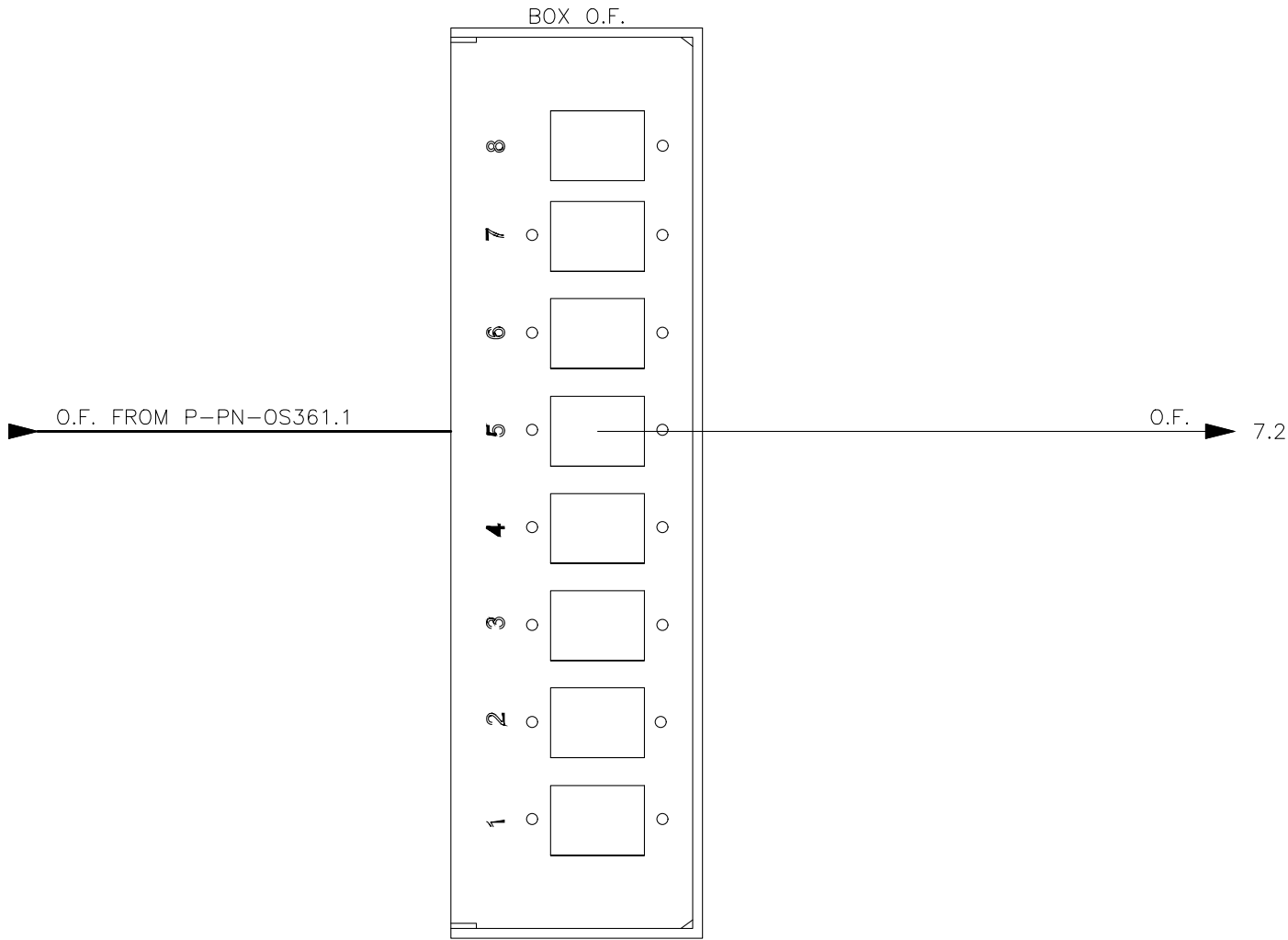
	POWER MAIN DISCONNECT SWITCH																FEEDER ID		
	1F+N+PE~50Hz 230V																RATED POWER Pn-RATED VOLTAGE Vn		
																	RATED CURRENT		

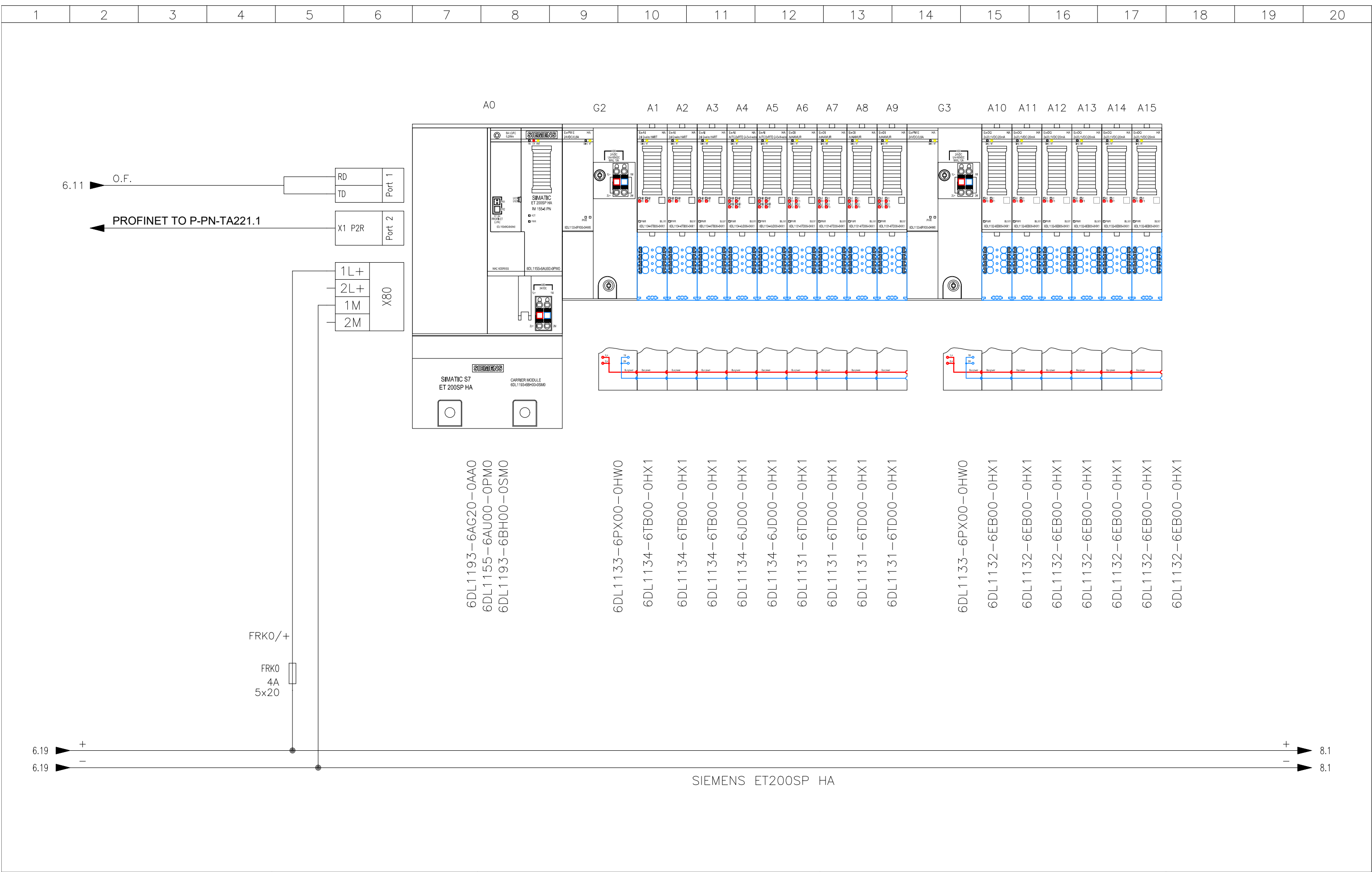




G1: TRIO-PS-2G/1AC/24DC/10 - Power supply unit PHORNIX CONTACT

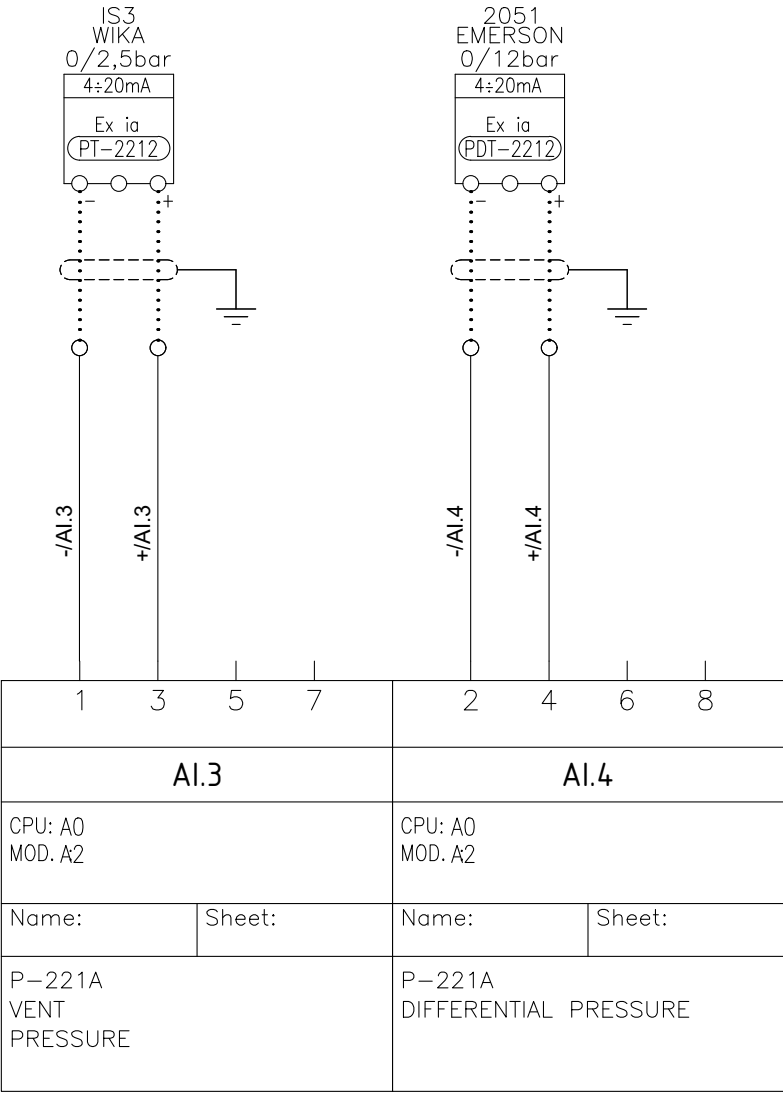
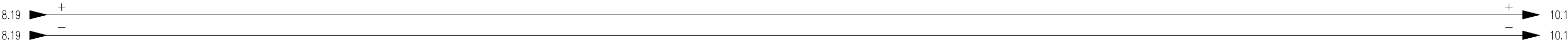
General Contractor : Manufacture : esa esaelctronicasrl.com		Project : P-PN-S221A.1 LNG PUMP SKID S-221A	Customer : ManaEnergy Managem Group	Object :		By :		
						Date : 26/03/2024		
Contract : IE24090			S/N : IE24090_SE01	Rev. 2	Data : 11/04/2025			
				Rev. 1	Data : 31/03/2025	Sheet : 5	of : 27	

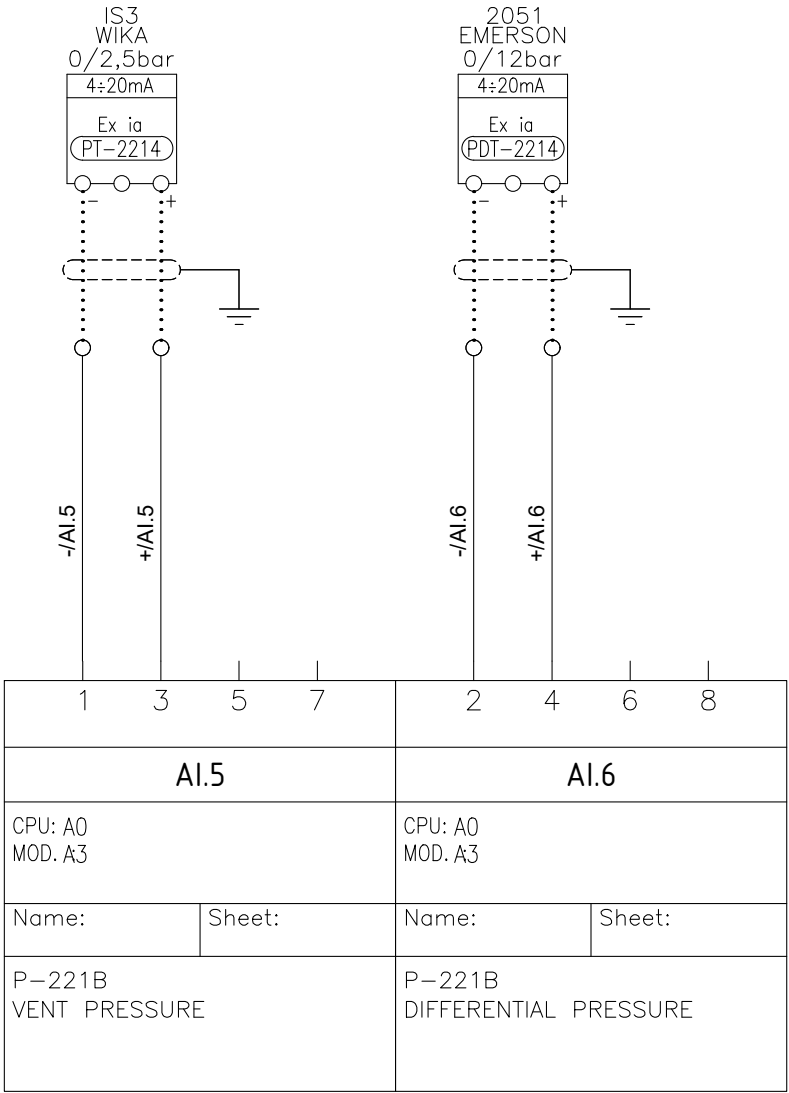
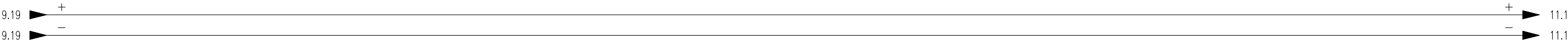


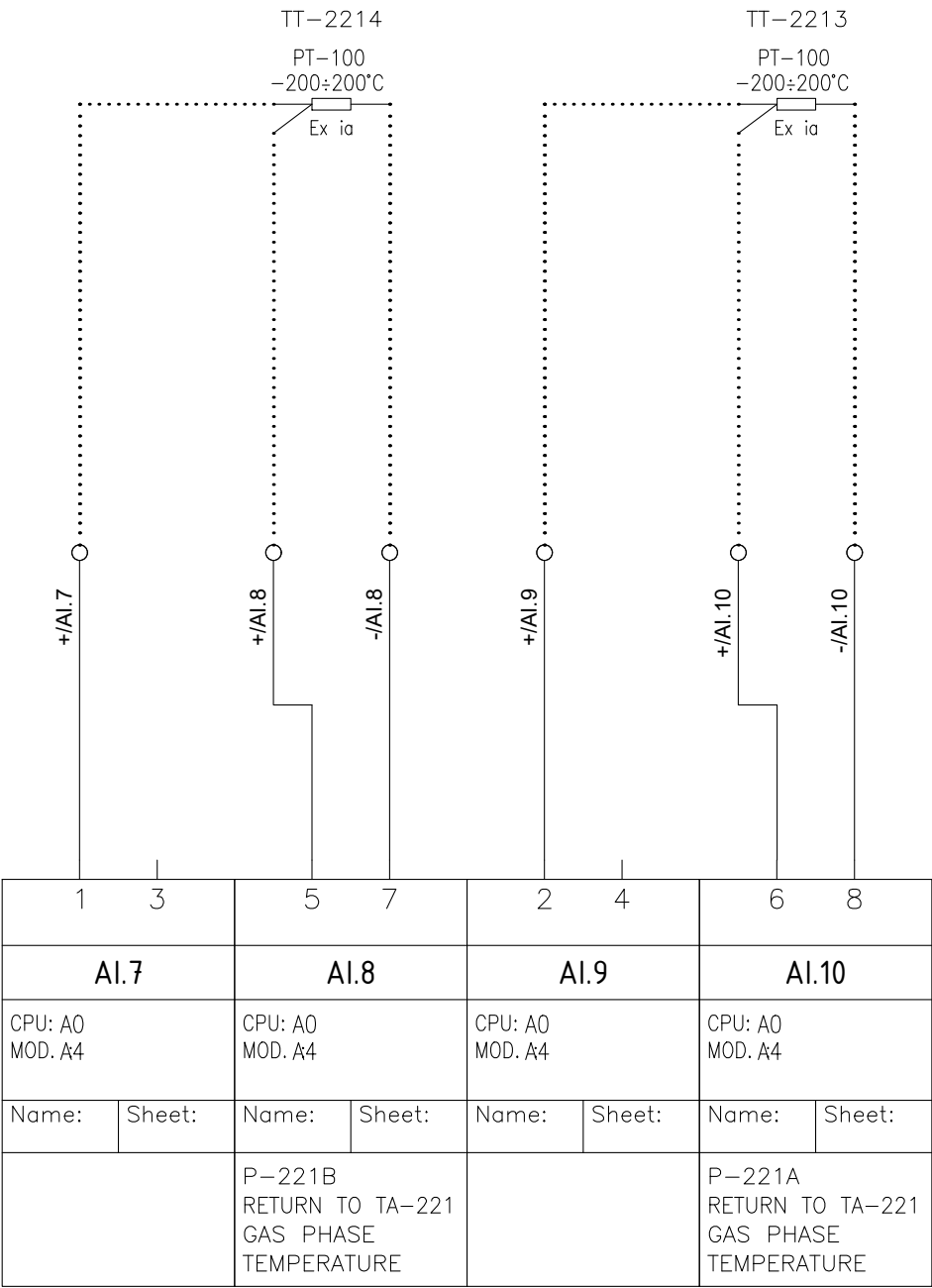
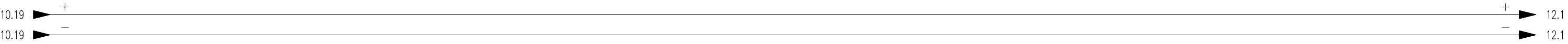


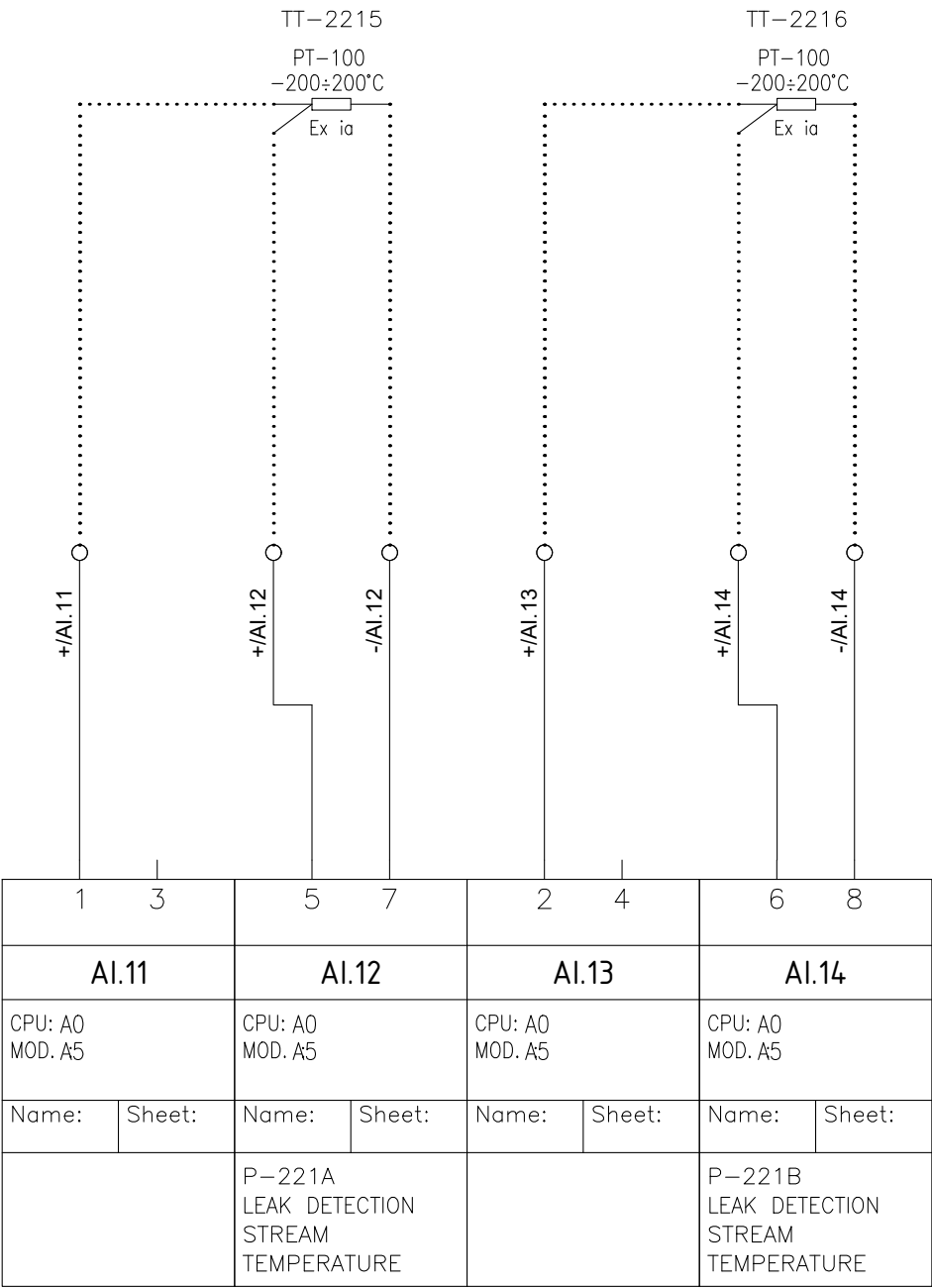
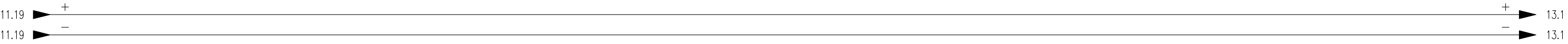
General Contractor : Manufacture : 		Project : P-PN-S221A.1 LNG PUMP SKID S-221A	Customer :  Contract : IE24090 S/N : IE24090_SE01	Object : Rev. 2 Data : 11/04/2025 Rev. 1 Data : 31/03/2025		By :	
						Date : 26/03/2024	Sheet : 7 of : 27

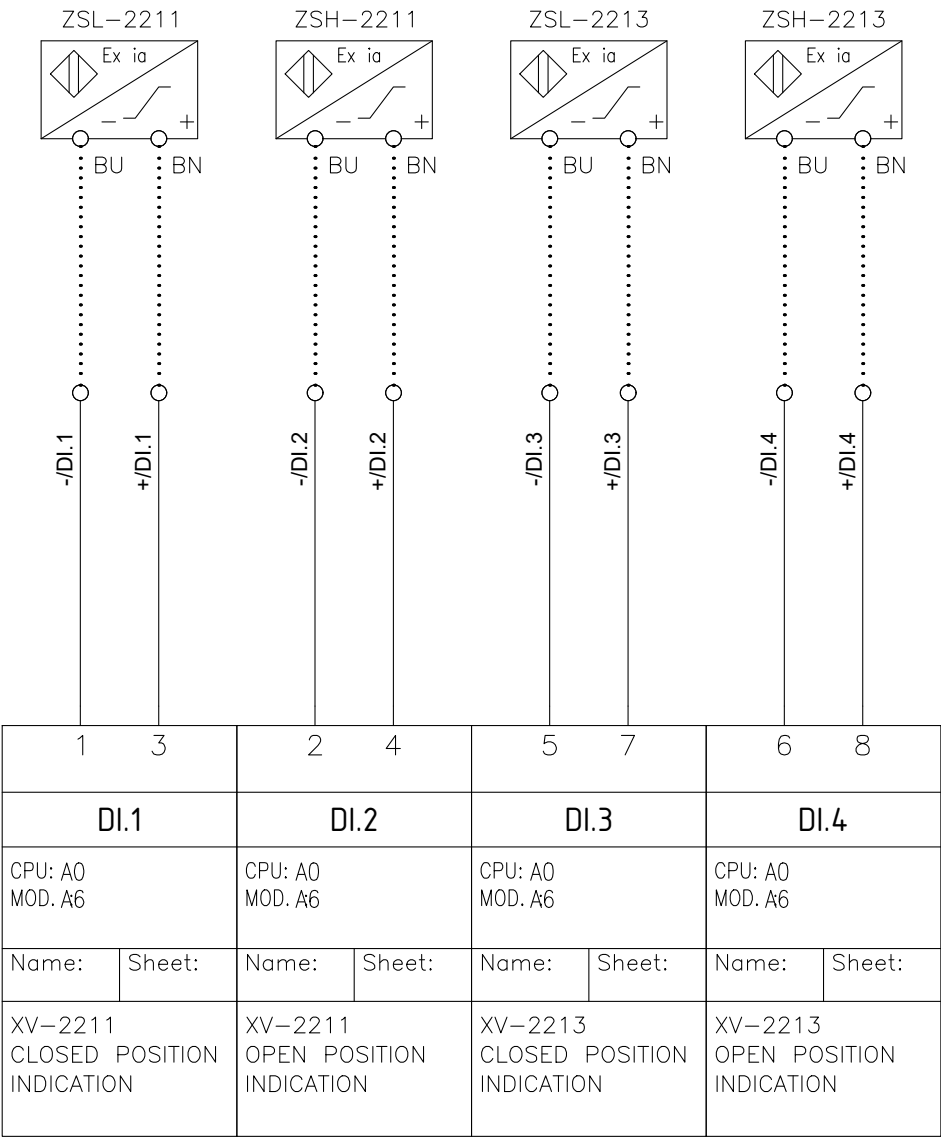
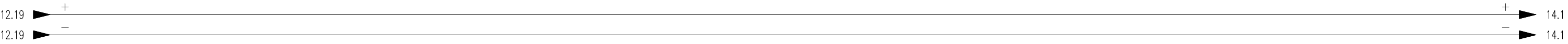




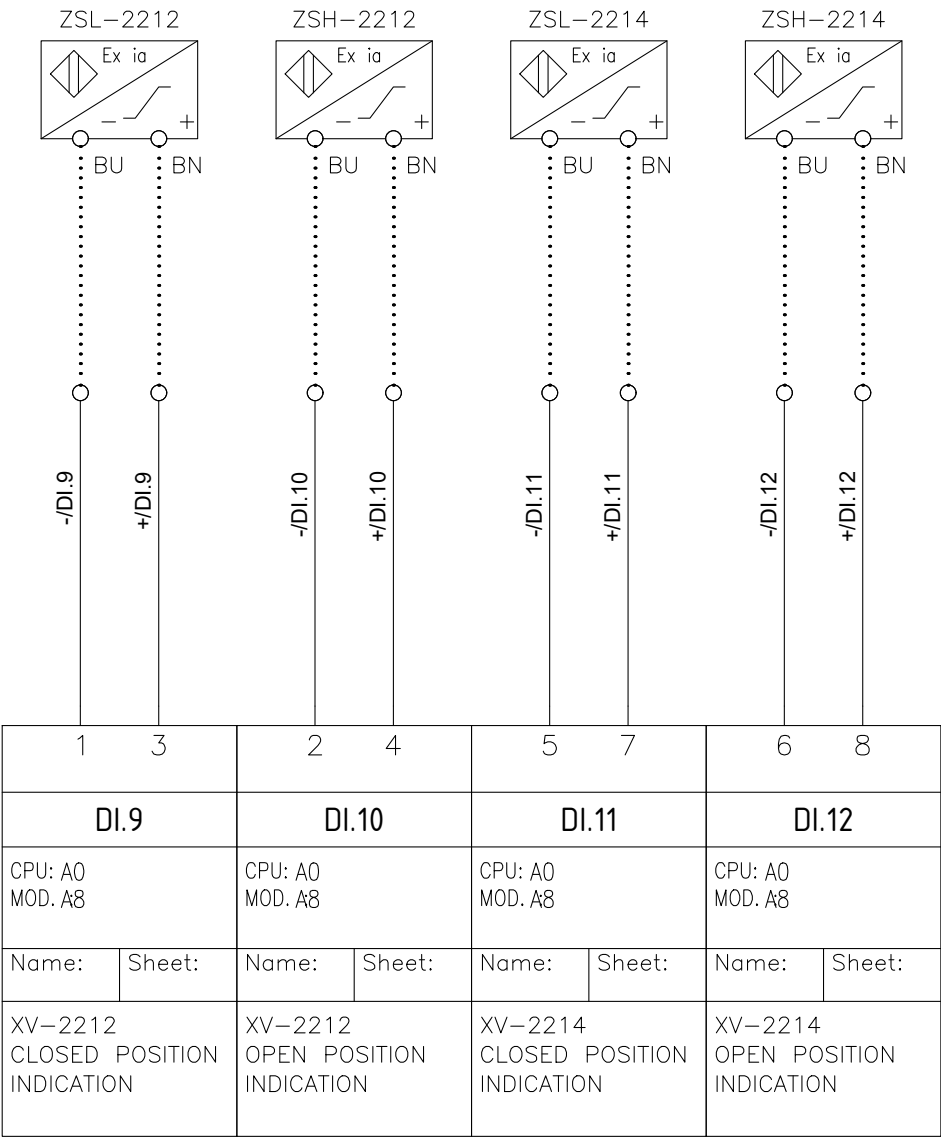
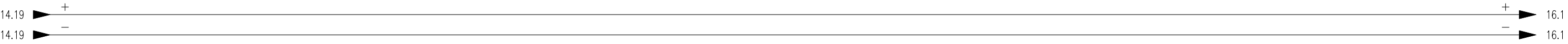


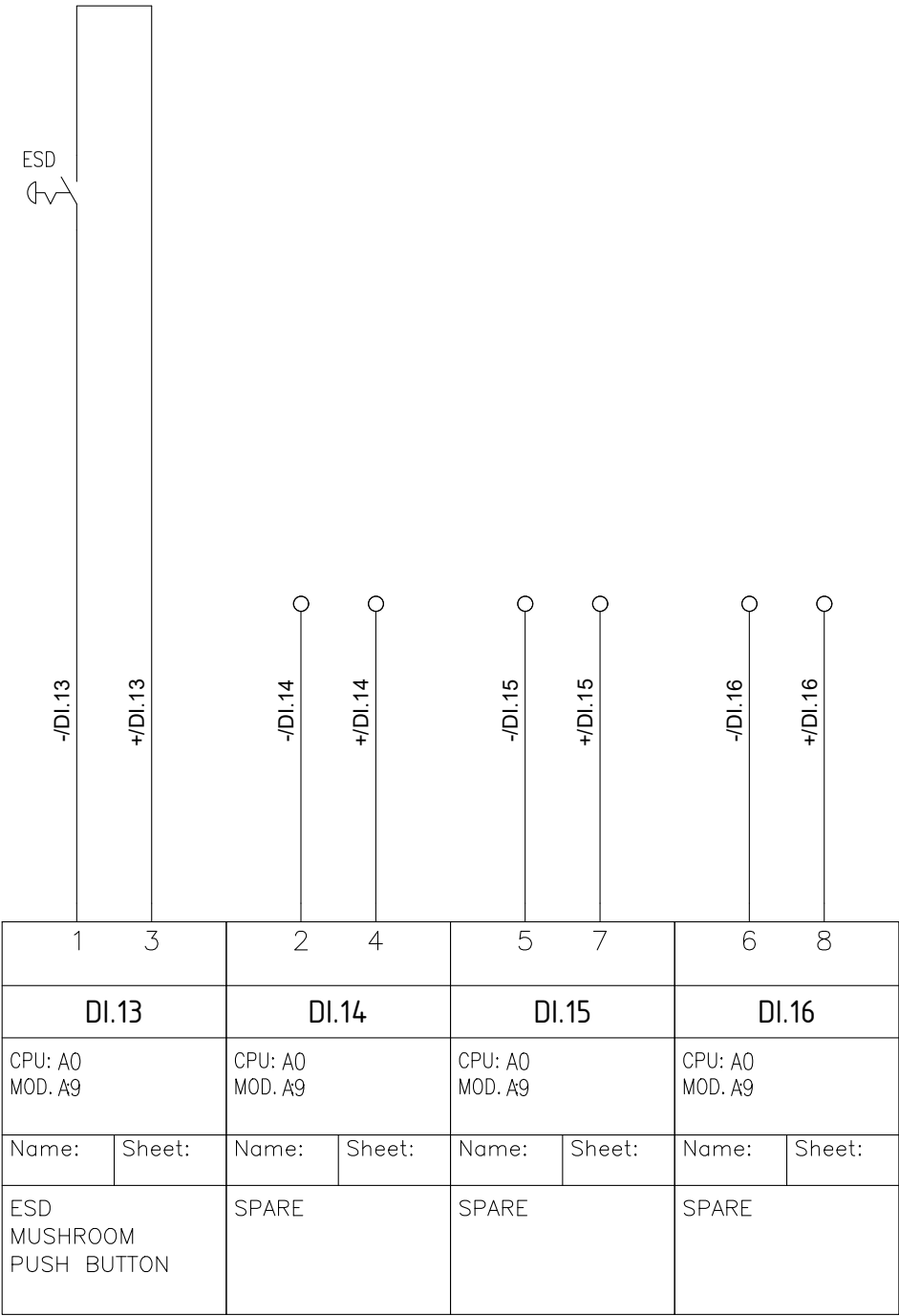
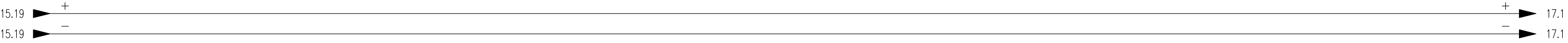


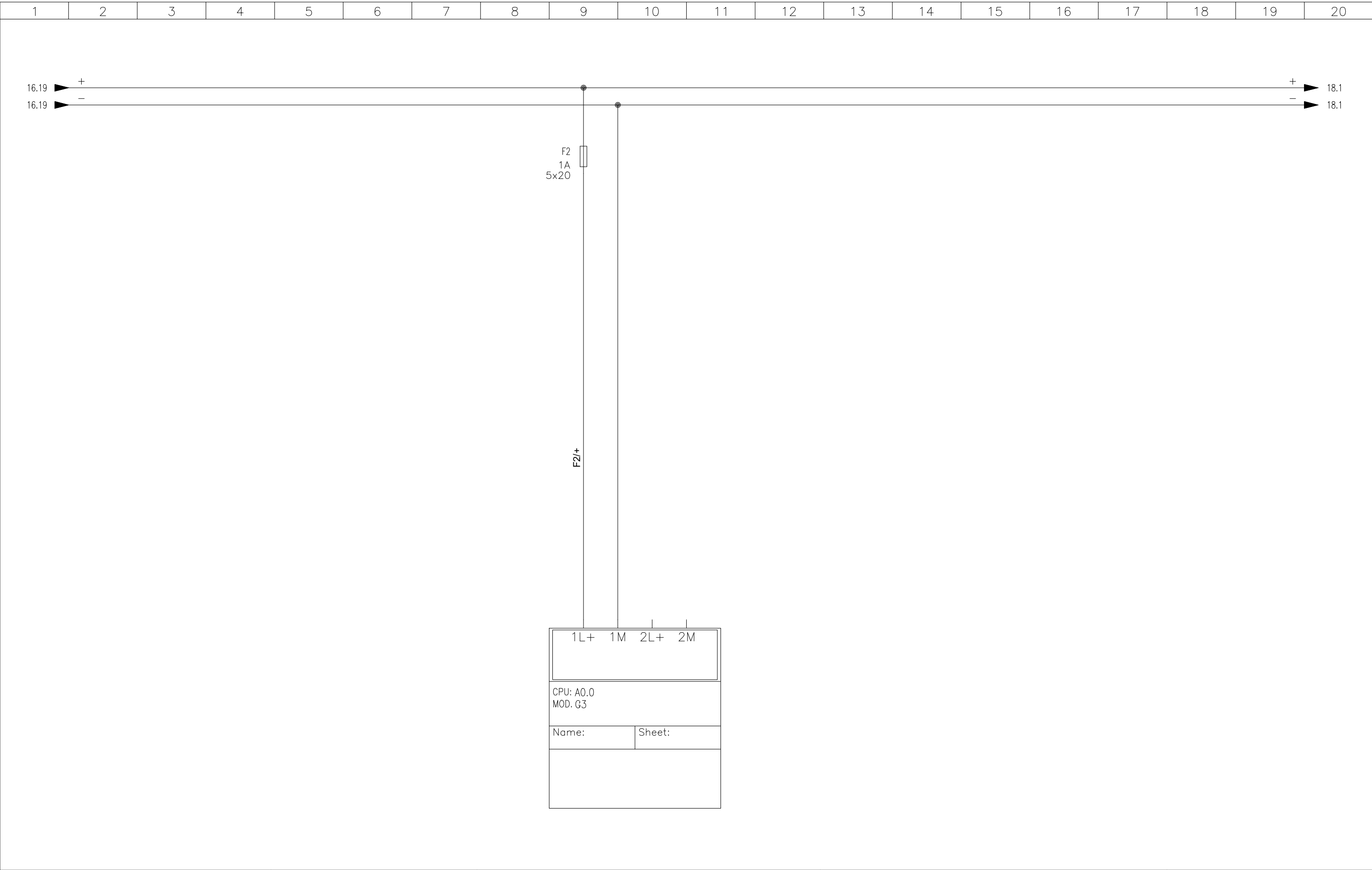


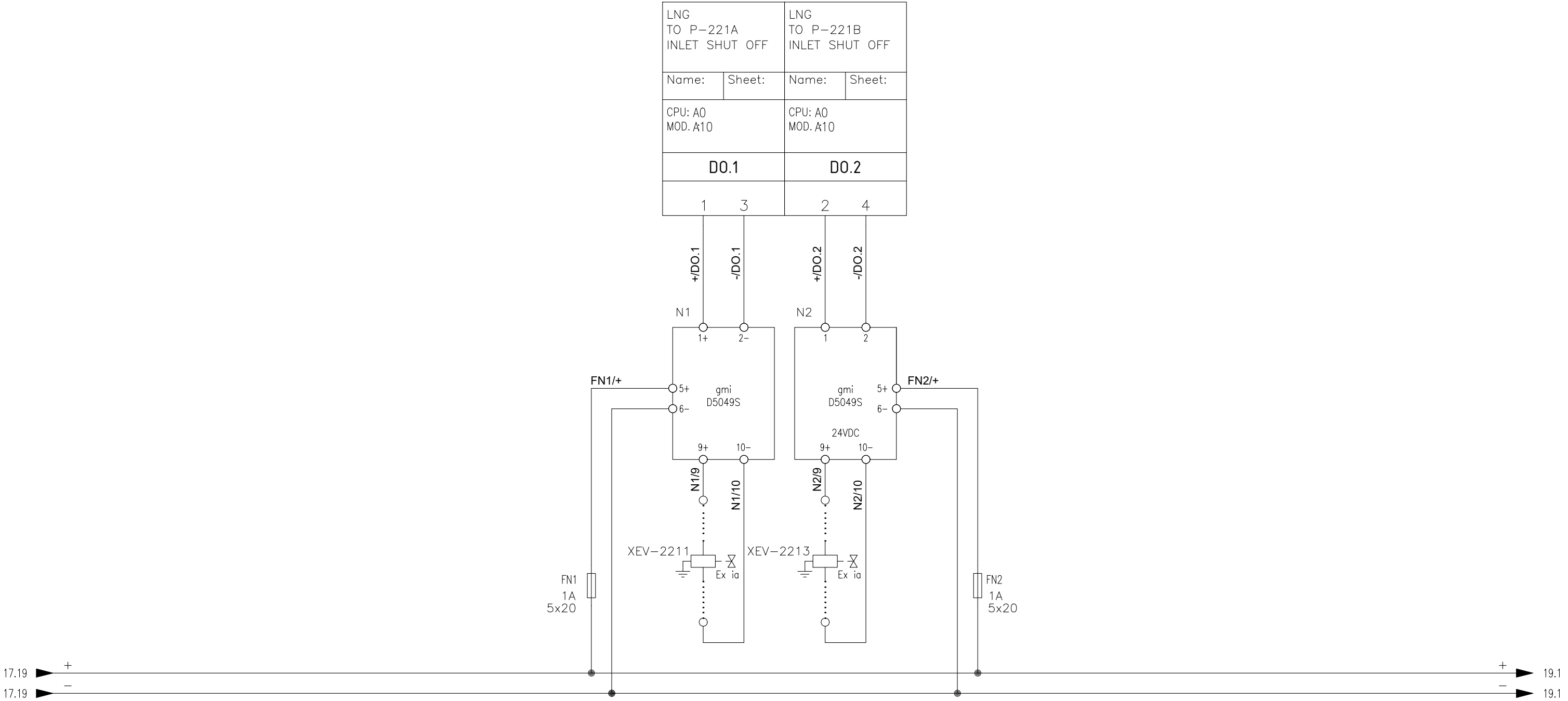


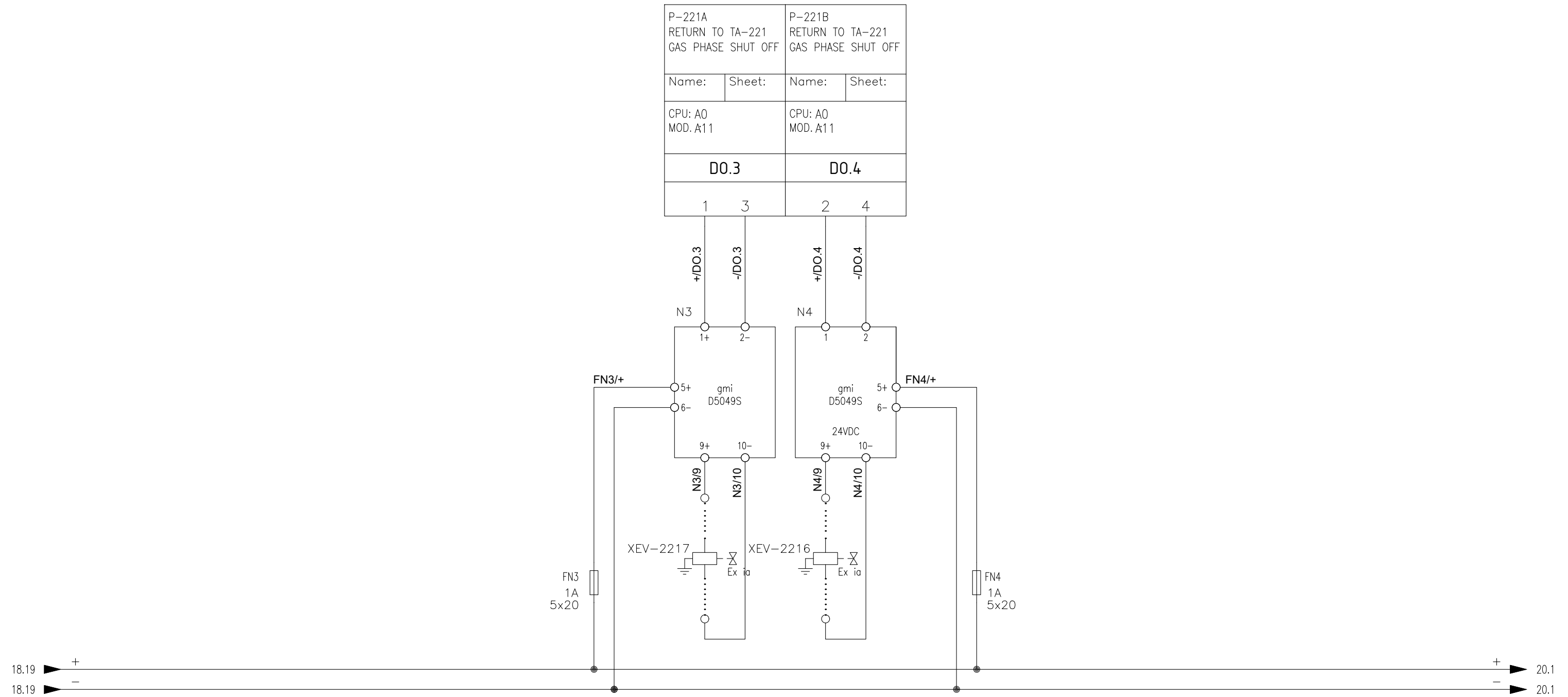


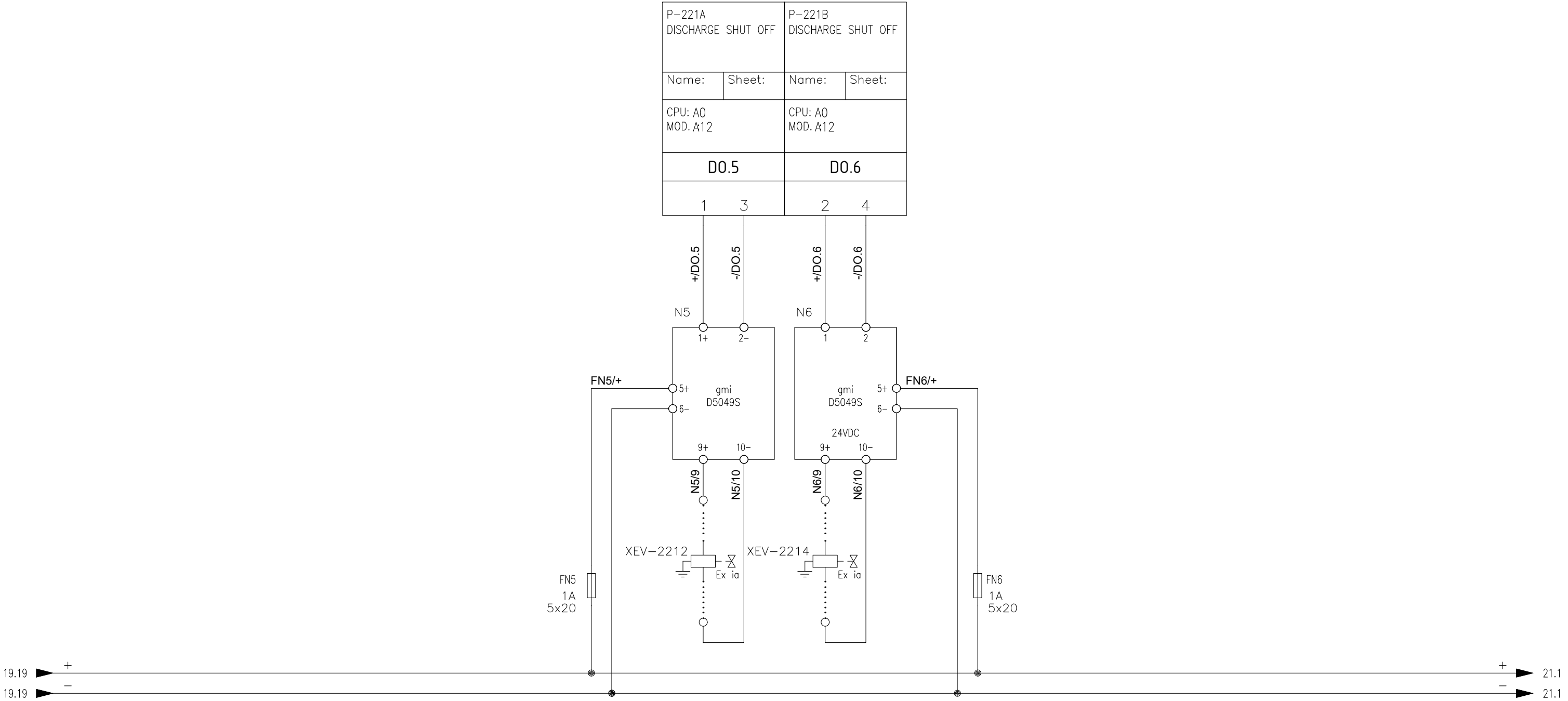


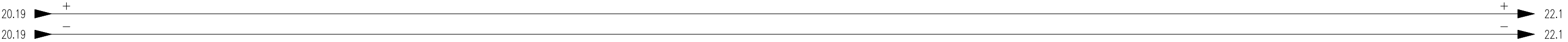
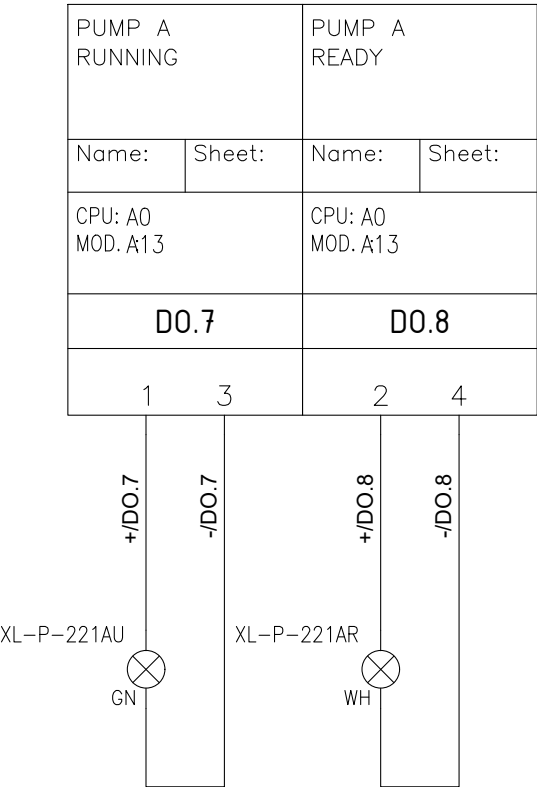


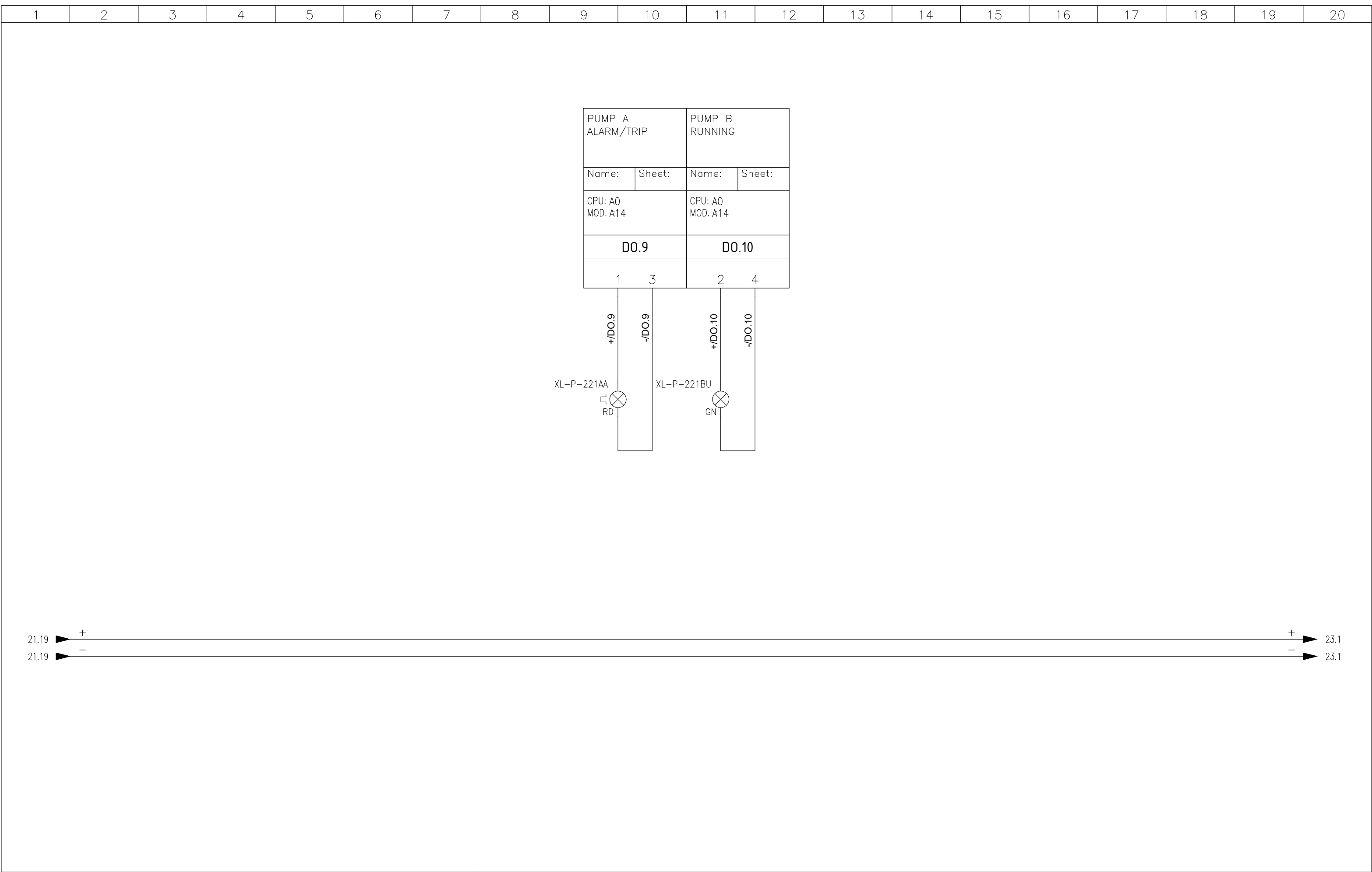


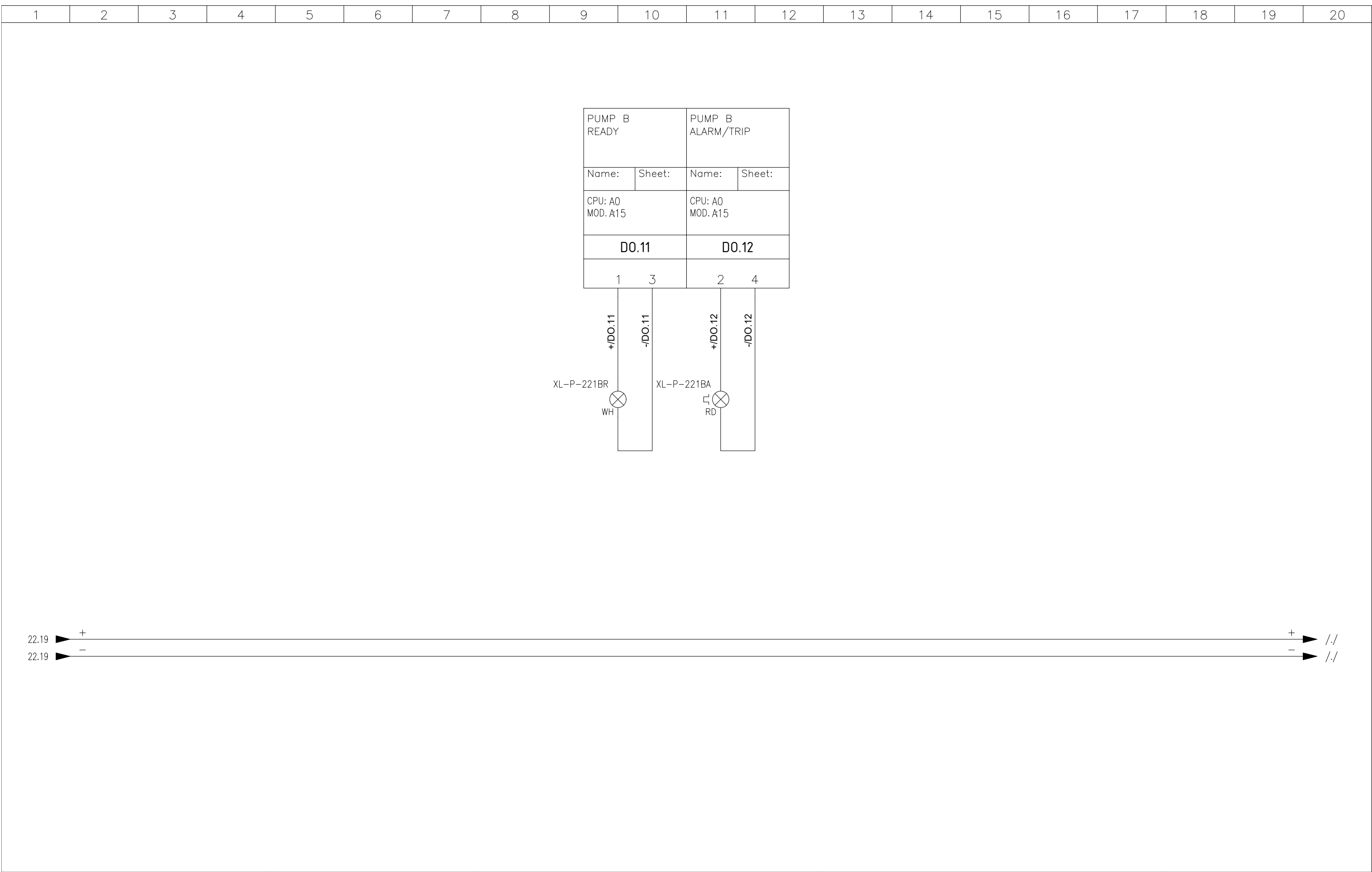






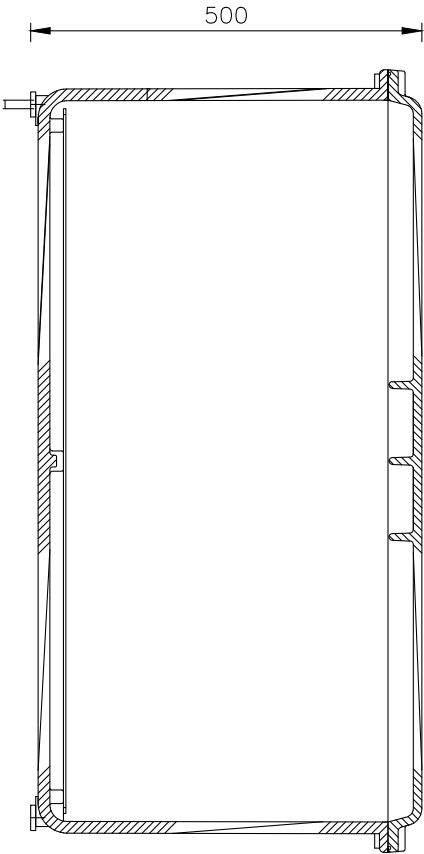
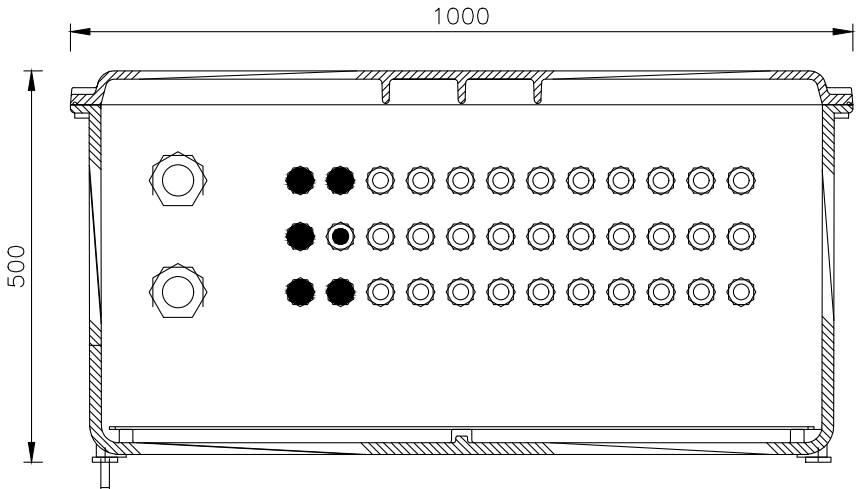
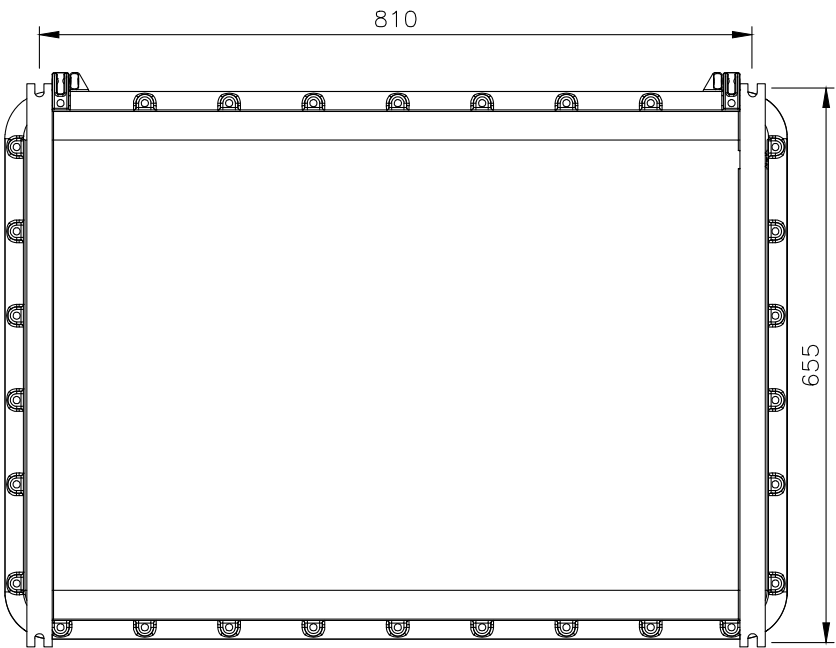
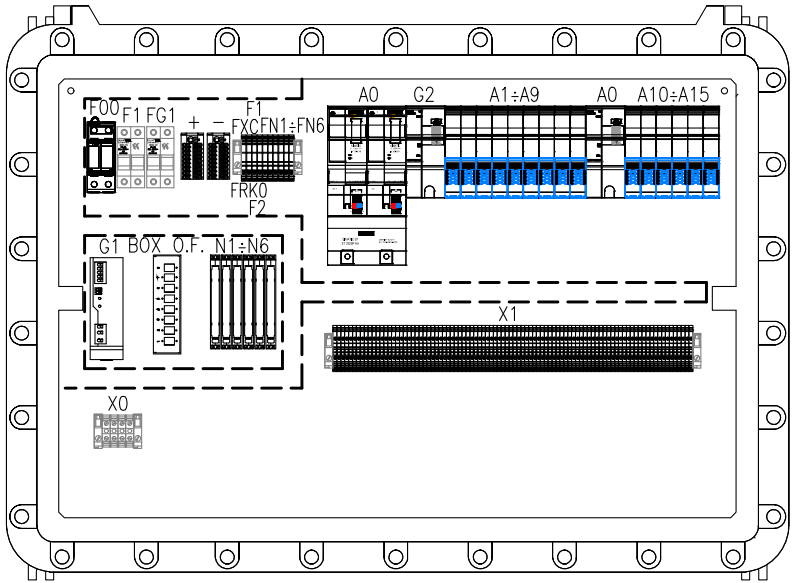
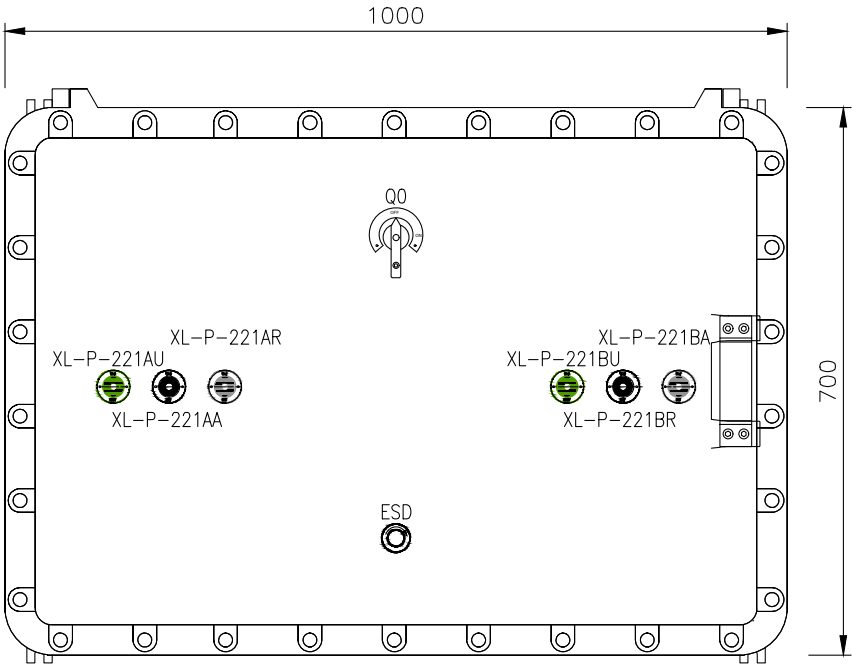


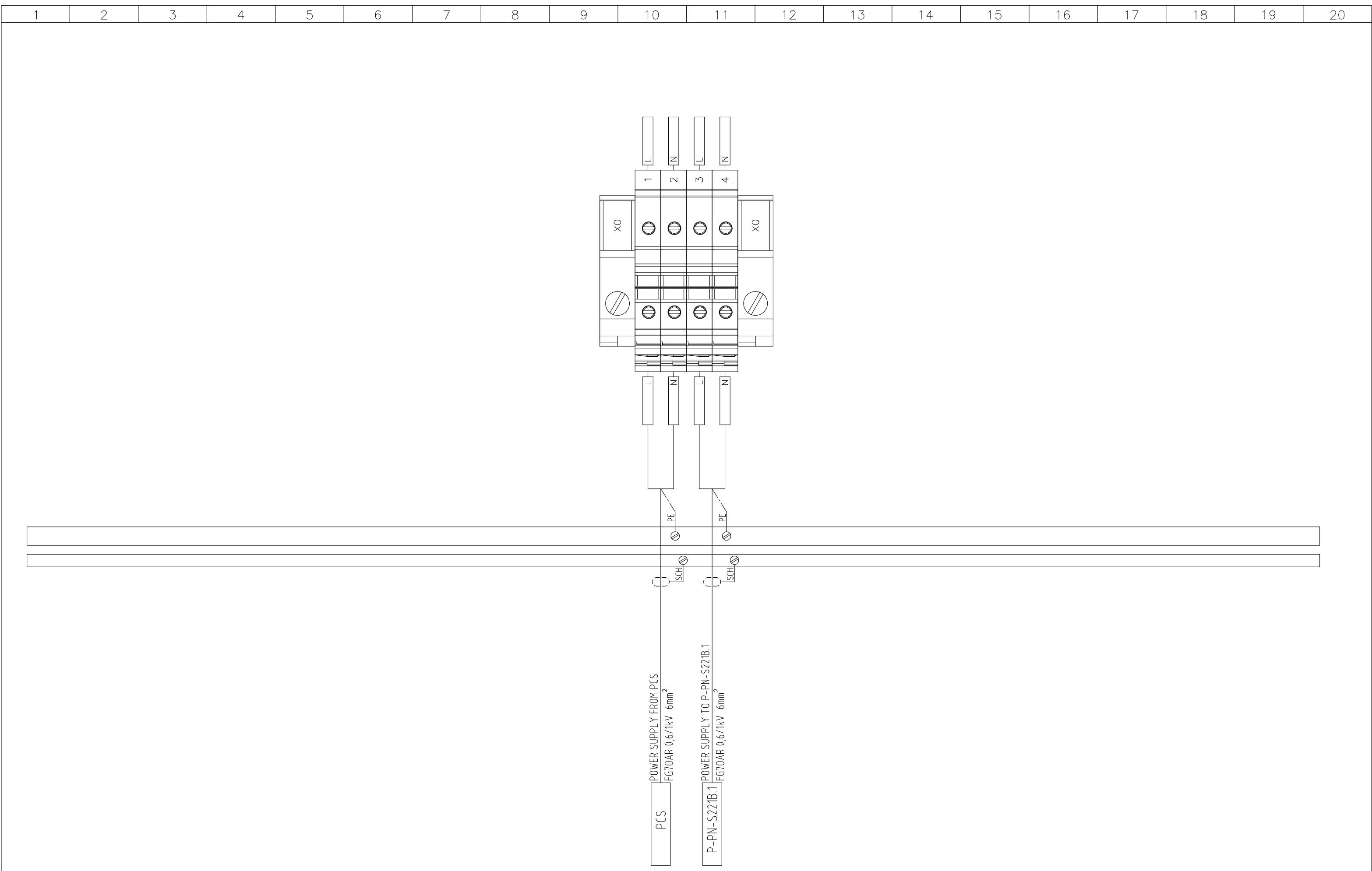




NOTE

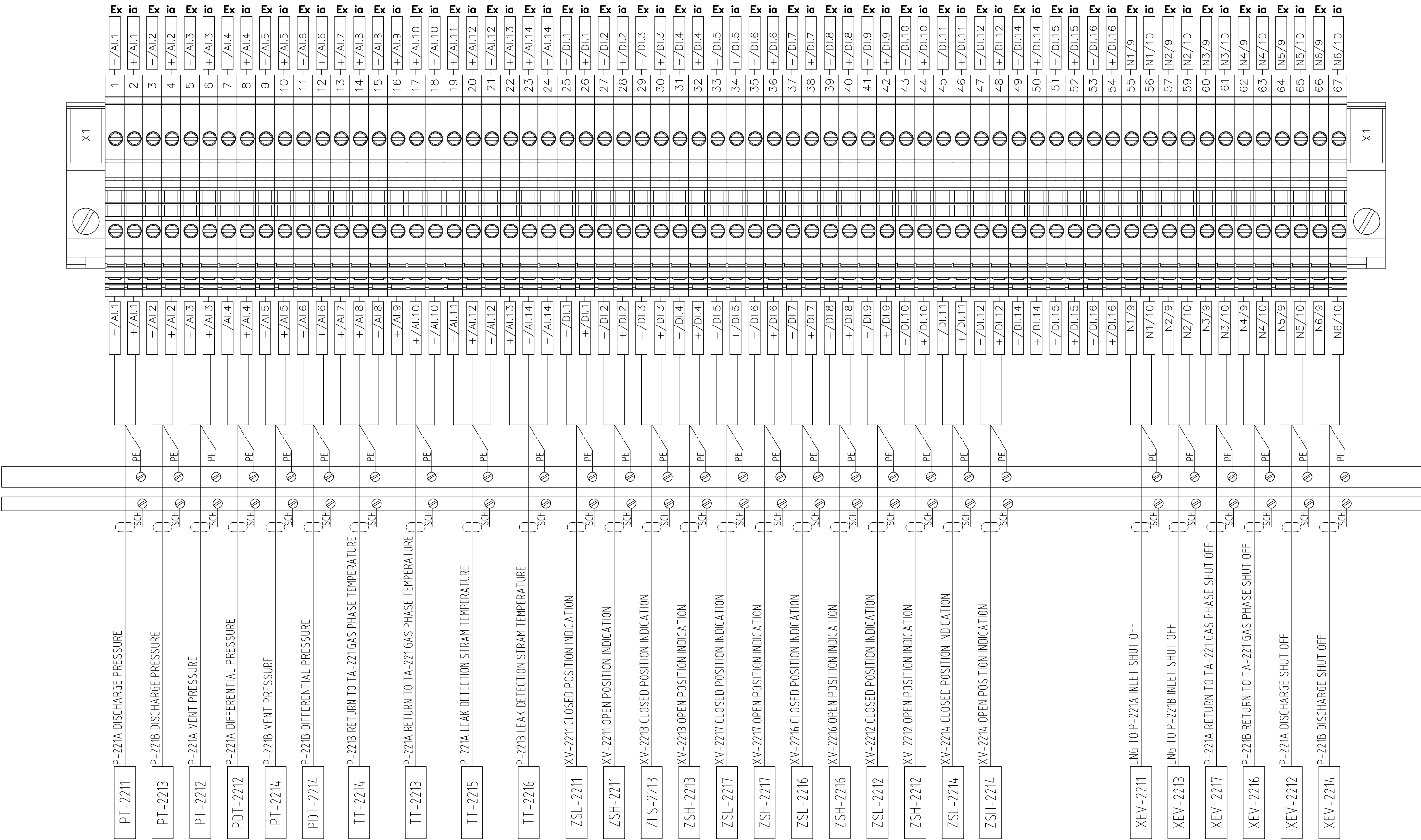
- Type : EJB-7
- Ex-d [ia] IIB T6 IP65 ATEX -20/+50°C
- Installation area is zone 2 IIA T1
- Weight 350kg
- Power dissipation 110W

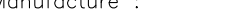




General Contractor : Manufacture : 		Project : P-PN-S221A.1 LNG PUMP SKID S-221A	Customer :  Contract : IE24090 S/N : IE24090_SE01		Object : Rev. 2 Data : 11/04/2025 Rev. 1 Data : 31/03/2025	By :	
						Date : 26/03/2024 Sheet : 26 of : 27	

INTRINSIC CIRCUIT INSIDE



General Contractor :		Manufacture : 	Project : P-PN-S221A.1 LNG PUMP SKID S-221A	Customer : 	Object :		By :	
					Rev. 2	Data : 11/04/2025	Date : 26/03/2024	
				Contract : IE24090	Rev. 1	Data : 31/03/2025	Sheet : 27	of : 27
				S/N : IE24090_SE01				